Universidade de Santiago de Compostela

Escola Técnica Superior de Enxeñaría
Master in Artificial Intelligence

Deliverable 2

Lab I4 & I5

Course:
Trustworthy and Explainable AI (TXAI)

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Overview

This document corresponds to the delivery **D2**, which includes **Assignments I4 and I5** of the course **Explainable and Trustworthy AI [P4251104] [2025/2026]**, within the **Master in Artificial Intelligence**.

The main objective of this work is to evaluate and ensure the **trustworthiness** and **fairness** of machine learning systems. While previous stages focused on interpretability, this delivery addresses the detection of systematic biases and the implementation of algorithmic strategies to mitigate discrimination.

In **I4** (Bias in the Lab), we examine the **robustness** of data and models using the **FAT-forensics** library. We focus on identifying **sampling bias** to determine if different subpopulations are equitably represented, and we evaluate **systematic performance bias** to ensure that predictive metrics remain consistent across demographic subgroups.

In **I5** (Fairness in the Lab), we focus on **algorithmic fairness** through the use of the **Fairlearn** library and **Aequitas** audit reports. We evaluate key metrics such as **Demographic Parity** and **Equalized Odds** to identify potential disparities. Furthermore, we apply and compare **mitigation strategies**—including **CorrelationRemover** and **GridSearch** constraints—to manage the inherent **fairness-accuracy trade-off** and ensure equitable outcomes in sensitive decision-making contexts.

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1 I4 — Bias in the Lab

1.1 Exercise I4.1

Problem

The function `fatf_dam.sampling_bias_grid_check` uses a default threshold of 0.8 to compute the pairwise sampling bias. Let us now modify this threshold to use a value of 0.5, and check the resulting output.

- Do you observe any change in reported results? How? Why?
- What does the given threshold account for? (you might want to check the corresponding API documentation)
- Given the defined group counts, what would be the minimal threshold that would cause the function to inform of existence of sampling bias?

Answer

Based on the execution of the modified code and the mathematical definition of the threshold, we can answer the questions as follows:

1. **Change in reported results:** Yes, there is a clear change in the output. By applying a threshold of 0.5, the disparity matrix is filled entirely with `False` values, and the text indicates that there is NO Sampling Bias reported for any pair of subgroups.

   Why it happens: A threshold of 0.5 means we are tolerating a massive 50% deviation from perfect parity (which would be a ratio of 1.0). In other words, the ratio between any two group sizes can freely fluctuate between 0.5 and 1.5 without triggering an alert. Since our worst-case ratio between groups is roughly 1.222, it now fits comfortably within this acceptable boundary, leading the algorithm to conclude that there is no bias.

2. **Meaning of the threshold:** The threshold accounts for the *tolerance to mathematical disparity* between the sizes of different subgroups. It determines how much the group counts can deviate from perfect equality (a ratio of 1.0) before being considered unfair or biased.

   The closer the threshold is to 1.0, the stricter the check becomes (demanding groups to be almost identical in size). Conversely, a lower threshold permits much more inequality before reporting an issue.

3. **Minimal threshold to detect bias:** To find the exact critical point, we must look at the most imbalanced pair of groups in our data: the smallest group (45 instances) and the largest group (55 instances).

   The maximum disparity ratio is:

$$\text{Ratio} = \frac{55}{45} \approx 1.222$$

The exact deviation from a perfectly balanced ratio (1.0) is 0.222. Therefore, the limit tolerance threshold is calculated around this boundary.

The minimal threshold is approximately **0.777**. If you set a threshold of 0.77 (which allows ratios up to 1.23), no bias is detected. However, if you set a threshold of 0.78 (which only allows ratios up to 1.22), our 1.222 ratio will break the limit, and the function **will** report the existence of sampling bias again.

1.2 Exercise I4.2

Problem

Similar to the previous `fatf_dam.sampling_bias_grid_check`, `fatf_accountability_models.systematic_performance_bias_grid`, you can use certain thresholds for this evaluation and answer the following questions:

- What is the default threshold applied in this case?
- Is the criterion for bias evaluation somehow different to the case of sampling bias seen in the previous section?
- Experiment with the code below and find a threshold that would result in the model to be qualified as containing systematic performance bias. Which groups would be involved?

Answer

Based on the analysis of the `systematic_performance_bias_grid` function and the calculated accuracy metrics for each sub-population, we can answer the questions as follows:

1. **Default threshold:** The default threshold applied by the `systematic_performance_bias_grid` function is **0.8** (following the standard 80% rule or four-fifths rule for disparate impact). Because the maximum deviation in our metrics does not exceed the 20% tolerance limit permitted by this threshold, the system does not detect any systematic performance bias by default.
2. **Criterion comparison (Sampling Bias vs. Performance Bias):** The mathematical criterion is exactly the same, as it relies on computing the disparity ratio between two values and checking if the maximum deviation exceeds the threshold tolerance. However, the *target* of the evaluation is completely different:
   - In the **Sampling Bias** section, the criterion evaluates the **Data** (it checks whether the number of instances is balanced across sub-populations).
   - In this **Systematic Performance Bias** section, the criterion evaluates the **Model** (it checks whether a specific predictive metric, such as accuracy, is equitable across the different sub-populations).
3. **Finding the triggering threshold:** To find the exact trigger threshold, we must calculate the disparity ratio between the best performing group (Group 0: 1.0 accuracy) and the worst performing group (Group 2: ≈ 0.909 accuracy).

The ratio is:

$$\text{Ratio} = \frac{1.0}{0.909} \approx 1.1$$

This indicates a deviation of 0.1 from the perfect equilibrium (1.0). Since the maximum deviation is 0.1, applying a threshold of **0.9** (which sets the maximum allowed deviation to exactly $1 - 0.9 = 0.1$) is enough to reach the boundary and flag the existence of Systematic Performance Bias.

The groups involved in this disparity are **Group 0** (the first subgroup: petal length ≤ 2.5) and **Group 2** (the third subgroup: petal length > 4.75). To make it completely clear, testing a threshold just below that limit, like 0.8999999999999, yields a matrix of all `False` values, proving that 0.9 is the exact breaking point.

1.3 Exercise I4.3

Problem

Execute now the code below, which makes use of the function `fatf.utils.metrics.subgroup_metrics.performance_per_subgroup` for similar purposes. In this case we train the classifier using less data, and use a different feature and splitting criterion.

- Which conclusions can you draw now?
- Execute the previous code again, but now select "Positive Predictive Value (PPV)" as performance metric. Do you observe any difference?
- What does "PPV" account for?
- Which is the role of the parameter "label index" in the `performance_per_subgroup` function? Does it make a difference changing it ("0", "1", or "2") if using "accuracy"? Does it make a difference changing it in case of using "PPV"? Why?

Answer

Based on the execution using the new split (sepal width, $x \leq 3$ vs $3 < x$) and comparing Accuracy with Positive Predictive Value (PPV), we conclude the following:

1. **Conclusions on subgroup performance:** The two confusion matrices show that the classifier behaves very differently across sub-populations. A key pattern is that *class 2 is never predicted* (the third predicted row/column is all zeros), while many true class-2 samples are absorbed into class 1 (e.g., the 33 and 17 entries linked to class-2 being assigned to class 1). In contrast, class 0 is classified almost perfectly in both groups. This difference in error distribution impacts overall performance: the accuracy is 0.60 for $x \leq 3$ and 0.75 for $3 < x$. Since the ratio between the lowest and highest subgroup performance hits the boundary of the 0.8 rule, the check correctly flags **systematic performance bias** between these two groups.

2. **Differences when using PPV:** Yes, switching from *accuracy* to *PPV* changes both what is measured and how results should be interpreted. Accuracy summarizes "overall correctness" mixing all classes, whereas PPV is *class-conditional* (one-vs-rest). For example, if we evaluate class 1:

   - Group $x \leq 3$: PPV (class 1) $= \frac{42}{42+33} \approx 0.56$
   - Group $3 < x$: PPV (class 1) $= \frac{8}{8+17} \approx 0.32$

   This indicates a much stronger disparity for class 1 under PPV than what was visible with accuracy alone. It highlights that the bias issue is mainly about the reliability of class-1 predictions (confusing class 1 vs class 2).

3. **Meaning of PPV:** Positive Predictive Value (PPV) measures the *precision* of a "positive" prediction. Mathematically:

$$\text{PPV} = \frac{TP}{TP + FP}$$

   In words: among all the samples the model predicted as the selected class, how many actually belong to that class?

4. **Role of the `label_index` parameter:** It selects which class is treated as the *positive* class when computing a binary/one-vs-rest metric (like PPV).

   - **Using Accuracy:** Changing `label_index` makes **no difference**, because accuracy evaluates overall correctness across all classes simultaneously.
   - **Using PPV:** Changing `label_index` makes a **big difference**, because the metric depends entirely on the chosen class. `label_index = 0` yields perfect PPV in both groups; `label_index = 1` exposes the heavy false-positive disparity; and `label_index = 2` is undefined/zero because the model makes no positive predictions for class 2.

1.4 Exercise I4.4

Problem

Below, we show how to use data density estimation with the FAT-forensics library to "measure" to what extent a prediction should be trusted. We choose one data point in a sparse region and another one in a dense region.

- Check the API documentation and explain what the "density score" actually represents.
- Given the previous results: Should we trust the model's output classifications?
- Is there any difference in this regard between the dense and sparse point classifications?

Answer

By evaluating the `fatf_density.DensityCheck` results alongside the kNN prediction probabilities, we can assess prediction trustworthiness:

1. **Meaning of the density score:** The density score is a *local data-density indicator* computed using k-nearest neighbours. It uses the distance from a point to its n-th closest neighbour as a proxy for how populated the surrounding region is. With normalization enabled (`normalise_scores=True`), the values are rescaled to the $[0, 1]$ range:
   - Values **near 0** indicate a **dense** area (many similar samples nearby).
   - Values **near 1** indicate a **sparse** area (few similar samples, neighbours are far away).
2. **Trusting the model's classifications:** We should be cautious. The density score warns us whether the model is operating in a well-supported region of the training data. For the sparse point, the probabilities are $[0, 1, 0]$ with a density score ≈ 0.948. For the dense point, the probabilities are $[0, 0.333, 0.667]$ with a density score ≈ 0.102. Trust cannot be based on probability alone; it must be contextualized by the density.
3. **Differences between the dense and sparse points:** Yes, there is a fundamental difference in how we interpret their predictions:
   - **Sparse point (Lower trust):** The point lies in a highly unpopulated region (score close to 1). Even though the model outputs "full confidence" (100% probability for class 1), this merely means the few nearest neighbours share the same class, but they might be very far away. The model is extrapolating, so the prediction should be taken with extra care.
   - **Dense point (Medium trust):** The point sits in a high-density region (score close to 0), meaning the prediction is strongly supported by nearby empirical data. However, the probability distribution is split (33% vs 66%). This suggests that while there is plenty of evidence, the point lies very close to a class overlap or decision boundary, so the final classification is still somewhat uncertain.

1.5 Exercise I4.5.a

Problem

A court uses COMPAS to assist in parole decisions. As you have seen, results from Aequitas' analysis show racial disparities in the calculated metrics. The judge requests a specific recommendation based on quantitative evidence. Your task is to analyze group-level outcomes and make a justified decision on which metric to prioritize when setting a decision threshold. Using the fdf table obtained from Aequitas extract the absolute values of FPR and FNR for groups African-American and Caucasian of the protected attribute race.

Answer

Ethnicity	**FPR**	**FNR**
African-American	0.448	0.280
Caucasian	0.235	0.477

Problem

The judge doesn't understand technical terms like FPR and FNR and wants a simple recommendation.

- Explain in plain language what it means to be a false positive (FP) and a false negative (FN) in the context of COMPAS predicting recidivism.
- Overall, considering both FPR and FNR, which group experiences the greater unfairness in the current predictions?
- Suppose you have to choose only one metric (FPR or FNR) to ensure parity across racial groups. Which one would you select and why?
- In the context of colon cancer prediction, what does it mean to be a false positive (FP) and a false negative (FN)? In this case, what would be the most critical metric?

Answer

1. **Explaining False Positives and False Negatives in COMPAS:**

   In plain language, a **False Positive (FP)** occurs when the system wrongly flags someone as "high-risk." This means a person is denied parole and kept in prison even though they would not have committed another crime if released. A **False Negative (FN)** is the opposite: the system flags someone as "low-risk," leading to their release, but they then go on to commit a new crime.

2. **Assessing Racial Unfairness:** African-Americans experience the greater systemic unfairness. The data shows they are nearly twice as likely as Caucasians (44.8% vs 23.5%) to be mistakenly labeled high-risk (False Positive). Conversely, Caucasian defendants who actually go on to re-offend are significantly more likely (47.7% vs 28.0%) to be mistakenly labeled low-risk and released (False Negative). This creates a "double burden" where minorities are punished more for crimes they won't commit, while others are shown more leniency for crimes they will commit.

3. **Choosing a Metric for Parity:** If forced to choose one, we would select the **False Positive Rate (FPR)** to ensure parity. In a legal context, the principle of "innocent until proven guilty" suggests that the harm of a False Positive—unjustly depriving a person of their liberty—is a more direct violation of civil rights. Given the massive disparity where African-Americans face an FPR of 44.8% compared to 23.5% for Caucasians, prioritizing parity in FPR ensures that one racial group isn't systematically over-incarcerated due to algorithmic error.

4. **Medical Context: Colon Cancer Prediction:**

   - **False Positive (FP):** The model predicts a patient has cancer when they are actually healthy. This causes high anxiety and leads to unnecessary, invasive tests like biopsies, but the patient's life is not in danger.

- **False Negative (FN):** The model predicts a patient is healthy when they actually have cancer. This is the **most critical metric** because the cancer remains untreated and may spread, leading to a potentially fatal outcome. In medicine, the cost of a False Negative (loss of life) far outweighs the cost of a False Positive (extra testing).

Problem

We have seen that some groups are more vulnerable than others. It is possible that combining these attributes may reveal new inequities that are not detected when each variable is considered separately.

- **Feature Engineering:** Create a new column that combines the `sex` and `age` variables (e.g., `sex_age`).
- **Aequitas Audit:** Perform an *Aequitas* audit using this newly created column as the protected attribute to evaluate intersectional fairness.
- **Disparity Metrics:** For the metrics **FPR** (False Positive Rate), **FNR** (False Negative Rate), and **FDR** (False Discovery Rate):
  - Print the calculated disparities for each combined group.
  - Use `summary_plot` and `disparity_plot` to visualize the results.
- **Reflection:** Analyze the output to determine if new inequities appear within these combined groups that were previously hidden when variables were analyzed in isolation.

Answer

Table 1: Aequitas Audit Results: Intersectional Sex and Age Attributes

Attribute	Value	FPR	FNR	FDR	FPR Disp.	FNR Disp.	FDR Disp.
sex_age	Female_25 - 45	0.294	0.408	0.438	0.852	1.111	1.171
sex_age	Female_Greater than 45	0.153	0.662	0.593	0.443	1.804	1.586
sex_age	Female_Less than 25	0.617	0.177	0.537	1.788	0.482	1.436
sex_age	Male_25 - 45	0.345	0.367	0.374	1.000	1.000	1.000
sex_age	Male_Greater than 45	0.172	0.557	0.436	0.498	1.519	1.165
sex_age	Male_Less than 25	0.514	0.273	0.316	1.490	0.744	0.844

Problem

If we only had information about the sex attribute for 'fpr' and 'fnr', does the sex_age attribute add new nuances?

What if we only had information about the age attribute?

What can you observe for the 'fdr' metric? Are there any new disparities?

Answer

- **Nuances added to the sex attribute:** The intersectional `sex_age` attribute reveals critical nuances hidden when looking at gender in isolation. For example, females under 25 experience a massive False Positive Rate of 61.7% (0.617), meaning they are highly over-penalized. Furthermore, older females (Greater than 45) suffer from a disproportionately high False Negative Rate of 66.1% (0.661). These extreme variations prove that risk is not distributed evenly within the global "Female" or "Male" categories.

- **Nuances added to the age attribute:** When only the age attribute is used, the `sex_age` combination adds essential depth by showing that within the "Less than 25" age group, the False Positive Rate is notably worse for females (61.7%) than for males (51.4%). Similarly, in the "Greater than 45" bracket, the False Negative Rate disparity is more pronounced for females (66.1%) than for males (55.7%). This indicates that the model's errors vary heavily by gender even when individuals belong to the exact same age group.

- **Observations for the 'fdr' metric:** The 'fdr' (False Discovery Rate) metric in the intersectional audit reveals significant new disparities. In particular, it shows that females in the "Greater than 45" category have a strikingly high FDR of 59.3% (0.593). This means that when the model predicts a high risk of recidivism for this specific intersectional group, it is incorrect nearly 60% of the time. High-risk predictions for older females are much more likely to be false alarms compared to other demographic groups, failing the parity test and exposing a unique vulnerability.

1.6 Exercise I4.5.b

Problem

The adult dataset is used to predict whether income exceeds $50K/yr based on census data. Load the adult for aequitas dataset you can find in the folder ADULT in testlib.

- Use Aequitas to audit this new dataset and obtain the disparities for the sensitive attributes gender, work_cat and age_cat as we've seen before. (Metrics: 'fpr', 'fnr' and 'fdr' — disparity_tolerance =1.25). Summarise it all in a summary plot.

- What disparities can you observe?

Answer

Based on the Aequitas audit, the model fails the fairness threshold (1.25) across all three attributes, systematically underestimating the income of marginalized groups:

- Gender: Females face a heavily inflated False Negative Rate (FNR) and lower False Positive Rate (FPR) compared to males. The model unfairly assumes women earn less.

- Age (age_cat): Younger adults (<25) experience severely inflated FNRs compared to middle-aged groups, meaning their high incomes are frequently missed.

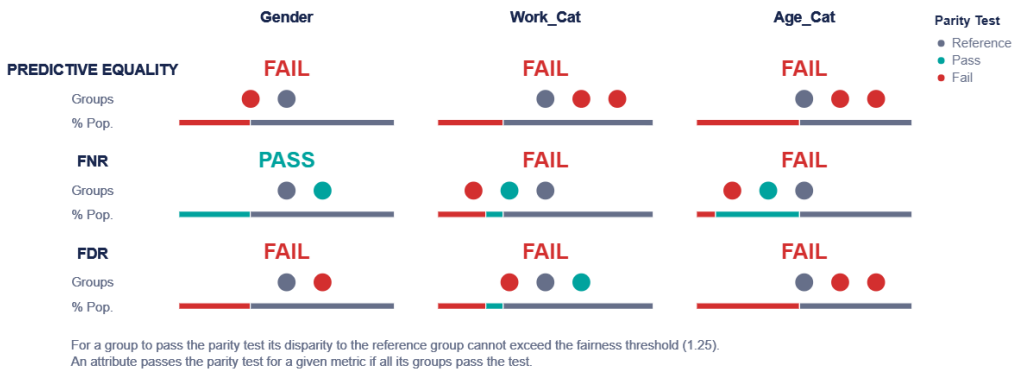

Figure 1: genderAge_4_5_b

- Work Class (work_cat): Lower-income job categories face higher FNRs, while categories like Federal Government or Self-Employed see higher FPRs compared to the Private sector baseline.

Problem

We've seen above that combining attributes may reveal new inequities. Combine the sensitive attributes gender and age in a new attribute. Audit the new dataset and get the summary plot with the same metrics and disparity tolerance as before.

- What can you observe for this new feature?
- Why is it not possible to calculate the relative disparity for some subgroups? What should be tuned to be able to calculate these disparities?
- If we were to tweak this metric, what possible implications would it have for the overall accuracy of the model?

Answer

Combining gender and age reveals intersectionality bias:

- Compounded Penalties: Individuals belonging to multiple marginalized groups (e.g., young females) face significantly worse error rates (e.g., higher FNR) than they did under single attributes.
- Concentrated Privilege: Favorable predictions heavily skew toward the most privileged intersectional group (e.g., middle-aged males).
- Hidden Biases Exposed: Subgroups that appeared fair when evaluated on a single attribute often fail the fairness audit when intersecting traits are evaluated together.

Relative disparity cannot be calculated if the reference group's metric is zero, which causes a division by zero error, or if a subgroup lacks enough data to compute the base metric. To resolve this, you should change the reference group to one with non-zero metrics or adjust the

feature categories by grouping smaller bins together to ensure sufficient data representation for all subgroups.

Tweaking fairness metrics to reduce disparities typically introduces a fairness-accuracy trade-off. By forcing the model to adjust its predictions—such as altering decision thresholds or reweighing data—to ensure equitable error rates across demographic subgroups, it deviates from the statistically optimal decision boundary for the general population. Consequently, while the model's predictions become fairer and less biased, its overall accuracy usually decreases.

2 I5 — Fairness in the Lab

2.1 Exercise I5.1

Problem

Modify the previous code with the aim of visualizing the fraction of people who earn over $50k regarding race. Then, plot correlations regarding age and race. Finally, remove correlations with `alpha=0.7` for `race_Black`.

Answer

To solve this exercise, we adapted the Fairlearn `MetricFrame` and `CorrelationRemover` tools to analyze and mitigate biases regarding the sensitive attribute *race*. The procedure and findings are structured as follows:

1. **Fraction of people earning over $50k by Race:** We grouped the dataset by the `race` feature and calculated the selection rate (fraction of positive cases). The quantitative results reveal significant disparities among different racial groups regarding high income:

   - **Asian-Pac-Islander:** 0.269 (26.9%)
   - **White:** 0.254 (25.4%)
   - **Other:** 0.123 (12.3%)
   - **Black:** 0.121 (12.1%)
   - **Amer-Indian-Eskimo:** 0.117 (11.7%)

   The maximum difference in selection rates between the highest and lowest groups is 0.152, and the minimum ratio is approximately 0.435. This clearly highlights a demographic disparity in the dataset's target variable.

2. **Original Correlations (Age and Race):** We isolated the `age` and `race` features, applied one-hot encoding to the categorical race data, and computed the initial correlation matrix. This baseline heatmap allows us to observe the natural linear dependencies between age and the different racial subgroups before any mitigation.

3. **Correlation Removal (`alpha=0.7` for `race_Black`):** We applied Fairlearn's `CorrelationRemover` explicitly targeting the `race_Black` feature. By setting `alpha=0.7`, we enforce a partial mitigation. The algorithm filters out the linear dependency by projecting the remaining features onto the null space of the sensitive feature, blending 70% of the perfectly

uncorrelated data with 30% of the original data. The resulting heatmap confirms that the correlation values between `race_Black` and the other variables decrease substantially, effectively obscuring the sensitive attribute from the rest of the feature space without completely destroying the original data structure.

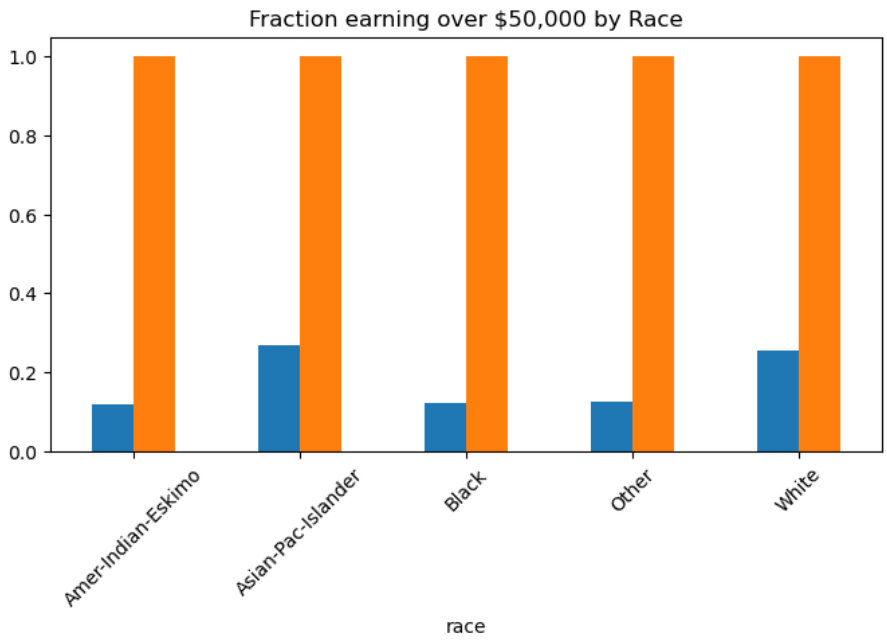

Figure 2: Fraction earning over $50,000 grouped by Race.

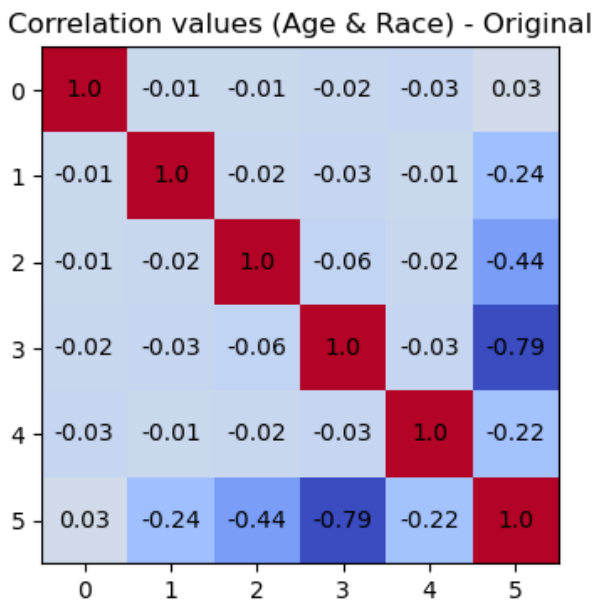

Figure 3: Original correlations (Age & Race).

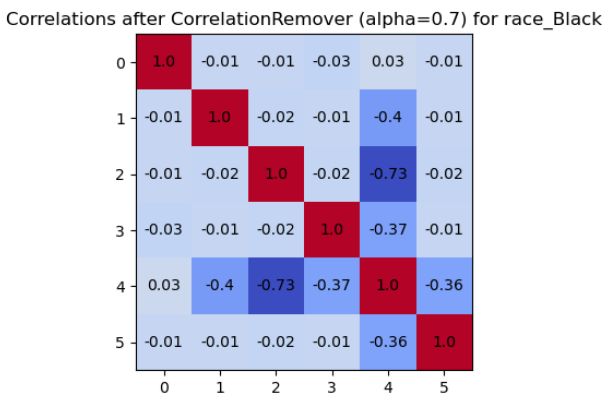

Figure 4: Correlations after `CorrelationRemover` ($\alpha = 0.7$) for `race_Black`.

2.2 Exercise I5.2

Analysis

The results reveal a severe bias favoring individuals with a Master's degree over High School graduates, despite removing education from the training data. There is a massive disparate impact, with the model predicting high income for nearly 63% of Master's holders compared to just 6.4% of HS graduates. Furthermore, the model exhibits a stark disparity in opportunity by correctly identifying 87% of actual high-earning Master's holders while missing almost 75% of actual high-earning HS graduates. The model is also far more likely to generously, yet incorrectly, guess a high income for a Master's holder (28% False Positive Rate) than for an HS graduate (under 3%). The slightly higher overall accuracy for HS graduates is merely a statistical illusion caused by the model aggressively predicting the negative class for this larger group. Ultimately, dropping the sensitive feature failed, as the model clearly reconstructed the education bias through other correlated proxy variables.

education	accuracy	selection_rate	true_positive_rate	true_negative_rate
HS-grad	0.859494	0.064051	0.253268	0.970641
Masters	0.808955	0.628358	0.870886	0.720000

education	false_positive_rate	false_negative_rate	count
HS-grad	0.029359	0.746732	3950.0
Masters	0.280000	0.129114	670.0

2.3 Exercise I5.3

— UNMITIGATED MODEL —

sex	accuracy	selection_rate
Female	0.928150	0.077264
Male	0.817233	0.253713

Overall Accuracy: 0.8541
Selection Rate Difference: 0.1764

— MITIGATED MODEL —

Overall Accuracy: 0.8321
Selection Rate Difference: 0.0032

sex	accuracy	selection_rate
Female	0.898868	0.149852
Male	0.798822	0.153062

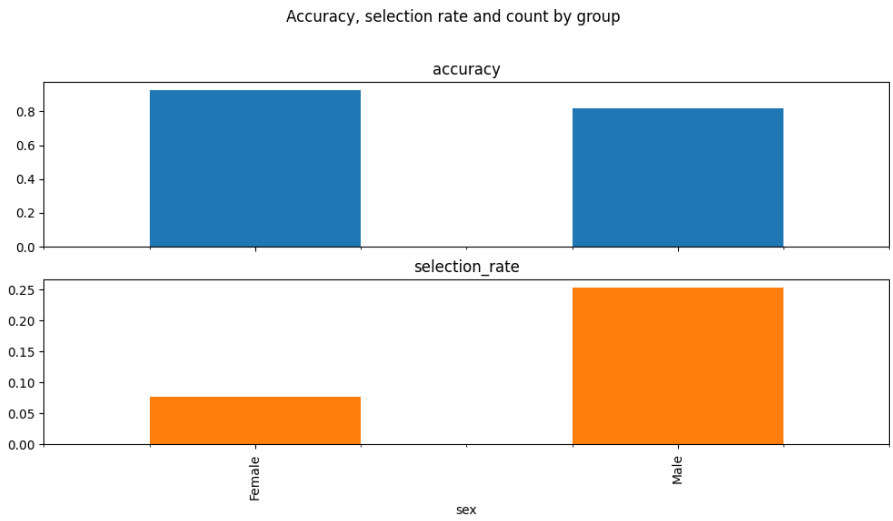

Figure 5: Normal Demographic

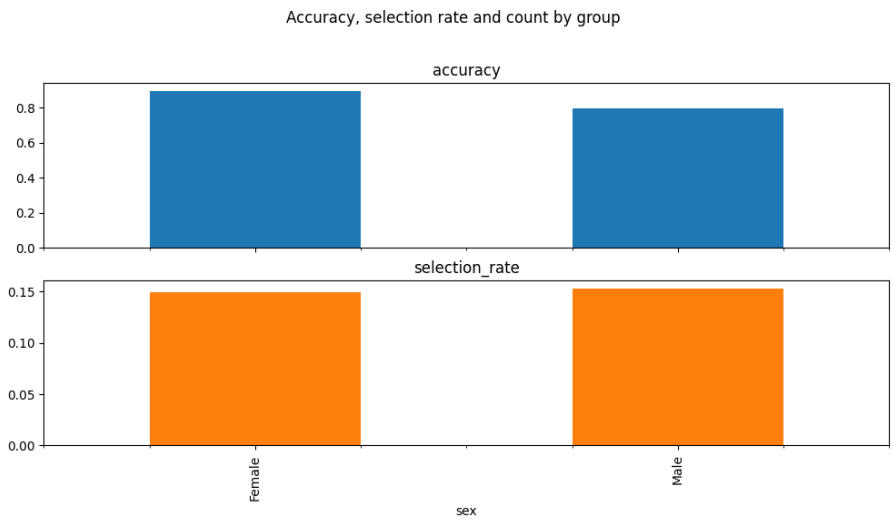

Figure 6: Mitigated Demographic

The results illustrate the classic fairness-accuracy trade-off. In the unmitigated model, there is a clear demographic bias, with males being predicted for high income at a significantly higher rate than females. After applying the Demographic Parity constraint, the mitigated model successfully balances these selection rates, ensuring both genders receive nearly equal opportunities for positive predictions. However, achieving this equitable outcome requires a mathematical compromise, resulting in a slight drop in the model's overall accuracy compared to the biased baseline.

2.4 Exercise I5.4 — Aequitas Fairness Report Analysis

Question 1

Select the adversarial debiasing report and answer: What does it mean Conditioned Demographic Parity in this context?

Answer

In the context of the Aequitas fairness report, **Conditioned Demographic Parity** goes a step further than standard Demographic Parity. It evaluates whether the model's predicted outcomes ("Pass" or "Not Pass") are distributed equally across different groups of a sensitive attribute (e.g., socioeconomic status), **conditioned on other legitimate variables or qualifications**.

Instead of simply demanding that High SES and Low SES students pass at the exact same blanket rate (which might ignore real academic differences), Conditioned Demographic Parity measures if students from different demographic backgrounds *who share similar legitimate characteristics* (such as identical previous grades or attendance records) receive the same proportion of positive outcomes. If achieved, it means the model is fair and does not favor a socioeconomic group when comparing students of equal academic merit.

Question 2

What is the significance of High socioeconomic status (SES) students showing a positive value while Low SES students show a negative value within this specific category?

Answer

Looking at the Conditioned Demographic Disparity charts for the "Pass" decision, High SES shows a positive bar while Low SES shows a negative bar.

The significance of this is that it highlights a clear bias or disparity in the outcomes:

- **Positive value (High SES):** This means that students with a high socioeconomic status are overrepresented in the positive outcome ("Pass") relative to the overall average.
- **Negative value (Low SES):** This indicates that students with a low socioeconomic status are underrepresented, receiving the "Pass" outcome at a lower rate than the baseline.

In short, it signifies that the model (or the historical data it learned from) favors the privileged group (High SES) and disadvantages the vulnerable group (Low SES) for successful academic outcomes.

Question 3

Do you notice any difference in performance and fairness metrics after applying the adversarial debiasing model?

Answer

No, there is absolutely no noticeable difference in either the performance or the fairness metrics after applying the adversarial debiasing technique.

- **Performance Metrics:** When analyzing the "Performance Plot," the metrics for Accuracy, Precision, Recall, ROC AUC, and F1 remain completely unchanged. In both the baseline ("No Mitigation") and the post-intervention ("With Mitigation") scenarios, all performance metrics sit at a perfect 1.0.
- **Fairness Metrics:** Similarly, the "Fairness Plot" shows zero changes. The Demographic Parity Ratio remains stagnant at approximately 0.68, and the Equalized Odds Ratio remains at 1.0 in both scenarios.

In conclusion, for this specific dataset and configuration, the adversarial debiasing mitigation technique had no impact. It neither harmed the model's predictive power nor helped reduce the existing demographic disparity.

Question 4

If you had to evaluate the model's performance by subgroups of a sensitive attribute, which metric do you think is not susceptible to the decision threshold? Why is this important?

Answer

Among the performance metrics typically evaluated in these fairness reports (such as Accuracy, Precision, Recall, ROC AUC, and F1), the metric that is not susceptible to the decision threshold is the **ROC AUC** (Receiver Operating Characteristic - Area Under the Curve).

Why is this important?

Most standard metrics (like Accuracy, Precision, and Recall) require a fixed decision threshold (e.g., setting a 0.5 cutoff) to classify a continuous probability into a discrete "Pass" or "Not Pass" outcome.

The ROC AUC is important because it measures the model's ability to rank positive instances higher than negative instances across *all* possible classification thresholds. When evaluating subgroups of a sensitive attribute, using ROC AUC is crucial for two main reasons:

1. **Assessing inherent predictive power:** It tells you if the model is fundamentally able to distinguish between positive and negative outcomes for a specific disadvantaged group, regardless of where the cutoff point is set.
2. **Identifying threshold-related bias:** If a model has low accuracy but a high ROC AUC for a vulnerable subgroup, it means the model's internal scoring is actually fine, but the chosen global threshold is misaligned for them. Conversely, a low ROC AUC indicates a deeper issue: the model's features or training are failing to properly represent and score that subgroup entirely.

Question 5

What value would indicate that group equity is achieved in the Demographic Parity and Equalized Odds metrics (think of it as a ratio)?

Answer

When evaluated as a ratio (such as the Demographic Parity Ratio or Equalized Odds Ratio), group equity is achieved when the value is **1.0** (or 100%).

This is because a fairness ratio is typically calculated by dividing the metric of the unprivileged group by the metric of the privileged group. A value of **1.0** means both groups are experiencing the exact same rates (e.g., the same selection rate for Demographic Parity, or the same true/false positive rates for Equalized Odds), indicating perfect parity. If the ratio is significantly less than 1.0 (for example, below the commonly accepted 0.8 threshold from the four-fifths rule), it indicates that the model is biased against the unprivileged group.

Question 6

Now go to the best-solution folder and answer: Do you observe any improvement or worsening in the model's performance after applying the mitigation technique?

Answer

Yes, there is a noticeable worsening in the model's overall predictive performance after applying the mitigation technique.

While the original model (without mitigation) achieved perfect performance scores (with Accuracy, Precision, Recall, ROC AUC, and F1 all at 1.0), the mitigated model shows a drop across these metrics. This illustrates the classic machine learning trade-off between fairness and performance: in order to achieve better equity and reduce bias across different socioeconomic subgroups, the model sacrifices a portion of its overall predictive accuracy.

Question 7

What does it mean that the Demographic Parity ratio has increased in this context? And Equalized Odds?

Answer

An increase in these fairness ratios (moving closer to the ideal value of 1.0) signifies a reduction in bias against the disadvantaged group (Low Socioeconomic Status).

- **Demographic Parity Ratio Increase:** This means that the disparity in overall selection rates is shrinking. In this context, it indicates that a higher proportion of Low SES students are receiving the positive "Pass" outcome, making their predicted pass rate more comparable to that of High SES students. The model's predictions are becoming more demographically balanced, meaning socioeconomic status is having less influence on simply receiving a positive prediction.
- **Equalized Odds Ratio Increase:** This implies that the model's error rates and accuracy rates are balancing out across both groups. Specifically, it means the model is becoming equally proficient at correctly identifying successful students (True Positive Rate) and equally prone to false alarms (False Positive Rate) for both Low and High SES students. Unlike Demographic Parity, Equalized Odds takes into account the actual ground truth, ensuring that genuinely qualified students from either socioeconomic background have the same equal opportunity to be correctly predicted to "Pass".

Question 8

You are comparing Technique A (improves Demographic Parity) and Technique B (improves Equalized Odds). Which would you prioritize in these scenarios? Justify your choice:

Answer

- **Recruitment System (Historical disadvantages):** I would prioritize **Technique A (Demographic Parity)**. When the training data reflects historical biases or systemic disadvantages, the historical "ground truth" labels are inherently flawed. Demographic Parity ignores these biased historical labels and ensures that the system selects candidates from all demographic subgroups at equal rates. This helps to ensure fair representation and breaks the cycle of historical discrimination.

- **Cancer Detection Model (Equal diagnostic error rates):** I would prioritize **Technique B (Equalized Odds)**. In medical diagnoses, the "ground truth" (whether a patient actually has cancer) is an objective biological fact, and the actual prevalence might naturally vary between demographic groups. Forcing Demographic Parity could lead to diagnosing healthy patients just to meet a quota. Equalized Odds is the correct metric here because it ensures that the model is equally accurate for all groups—meaning no specific patient group suffers from a disproportionately higher rate of missed diagnoses (false negatives) or false alarms (false positives).

Question 9

To what extent is the trade-off between performance and equity acceptable?

Answer

The acceptability of the trade-off between model performance and equity is not a strictly mathematical question, but rather a context-dependent socio-technical and ethical decision.

The extent to which this trade-off is acceptable depends heavily on the domain, the application, and the potential harm caused by biased predictions:

- **High-Stakes Applications:** In areas like criminal justice, healthcare, lending, or education (such as the academic risk model analyzed in this lab), equity is paramount. A drop in overall performance is generally an acceptable and necessary price to pay to ensure that vulnerable or historically marginalized groups are not disproportionately harmed or unfairly denied opportunities.

- **Low-Stakes Applications:** In domains like targeted advertising or product recommendations, a slight demographic skew might not cause significant allocative or representational harm. In these cases, stakeholders might prioritize overall accuracy and business metrics over perfect demographic parity.

Ultimately, deciding the acceptable threshold requires involving domain experts, ethicists, and affected communities to balance institutional goals with human rights and fairness principles.